

week-2

Day 1

Tedd

- Went through the basics of p5live again
- Showed us how to create and use snippets
- Experimented with the audio snippet

```
function setup() {
  createCanvas(windowWidth, windowHeight)

  setupAudio(true) // global vars
  // a5.ease = .075 // set easing
  setupAudio(true) // global vars
  // a5.ease = .075 // set easing
}

function draw() {
  //ctrl shift A = audiorecative snippet
  updateAudio()
  //Fade Out hinzugügen mit background
  background(0, 0, 255, 15)
  fill(255, 0, 0)
  //Kreis der von links nach rechts geht und auf Audio reagiert
  circle(frameCount * 10 % width, height / 2, ampEase * 10)
}
```


- Created a Webcam snippet

```
// {"P5LIVE":{"name":"webcam snipped used","mod":1777910742845}}

let capture, scl = 1,
    rot = 0

function setup() {
  createCanvas(windowWidth, windowHeight)

  createCanvas(windowWidth, windowHeight)
```

```

    capture = createCapture(VIDEO)
    capture.size(320, 240)
    capture.hide() // hide raw camera
    imageMode(CENTER)
  }

  function draw() {
    //option shift s = snippet bearbeiten
    //ctrl shift s = snippet einfügen
    scl = 2
    image(capture, mouseX, mouseY, capture.width * scl, capture.height
    * scl)

  }

```


- Showed us how to use snippets from online libraries

```

// {"P5LIVE":{"name":"walker snippet","mod":1777912133985}}

//library einbetten mit let libs tab
let libs =
['https://tetunori.github.io/BMWalker.js/dist/v0.6.1/bmwalker.js']

const bmw = new BMWalker();

function setup() {
  createCanvas(windowWidth, windowHeight)
  background(0, 0, 255)
}

//code aus dem snippet an richtigen Ort kopieren
//parameter nach Wunsch anpassen
function draw() {
  clear()
  //Grösse des Walkers
  const walkerHeight = 600;
  const markers = bmw.getMarkers(walkerHeight);

  //Position des Walkers bestimmen
  translate(width / 3, height / 2)
  markers.forEach((m) => {
    //Grösse der Kreise bestimmen
    circle(m.x, m.y, 80);
  });
}

```


Day 2

Alper

- Showed us how to create basic sliders
- Started with one slider and expanded it also with buttons

```
// {"P5LIVE":{"name":"slider 01","mod":1777973807214}}

let slider, checkbox, button, colorPicker, dropdown, input,
sliderText;

function setup() {
  createCanvas(windowWidth, windowHeight);

  //checkbox
  checkbox = createCheckbox('Show Form', true);
  checkbox.position(width - 400, 20);

  //slider1
  //first value = start Grösse des Kreises
  // second value = max Grösse des Kreises
  slider = createSlider(50, width - 100, 200);
  slider.position(width - 400, 60);
  //slider2
  slider2 = createSlider(50, height - 100, 200);
  slider2.position(width - 400, 80);
}

function draw() {
  background(220);

  if(checkbox.checked()) {
    //slider1 value bestimmt breite, slider 2 value höhe
    ellipse(width / 2, height / 2, slider.value(),
    slider2.value());
  }
}
```



```
// {"P5LIVE":{"name":"slider 01","mod":1777974610291}}

let slider, checkbox, button, colorPicker, dropdown, input, sliderText;
```

```

let bgColor;
let positionDOM;

function setup() {
  createCanvas(windowWidth, windowHeight);
  positionDOM = width - 400

  //checkbox (Erstellung Checkbox / Position)
  checkbox = createCheckbox('Show Form', false);
  checkbox.position(width - 400, 20);

  //sliders (Erstellung Slider / Position)
  // 50 = Startpunkt / height-100 = max Grösse
  slider = createSlider(50, height - 100, 400);
  slider.position(width - 400, 60);

  // Button
  button = createButton('Random Background');
  button.position(positionDOM, 100);
  button.mousePressed(() => {
    bgColor = color(random(255), random(255), random(255));
  });
  bgColor = color(220);

  // Color Picker
  colorPicker = createColorPicker('#ff0000');
  colorPicker.position(positionDOM, 140);
}

//
function draw() {
  background(bgColor);
  fill(colorPicker.value());

  if(checkbox.checked()) {
    ellipse(width / 2, height / 2, slider.value(), slider.value());
  }
}

```


- Created our own basic GUI

```

// {"P5LIVE":{"name":"GUI","mod":1777976055921}}

let slider, checkbox, button, colorPicker, dropdown, input,
sliderText, buttonText;
let bgColor;

```

```

let colorText;
let positionDOM;

function setup() {
  createCanvas(windowWidth, windowHeight);
  positionDOM = width - 400

  //checkbox (Erstellung Checkbox / Position)
  checkbox = createCheckbox('Show Form', false);
  checkbox.position(width - 400, 20);

  //sliders (Erstellung Slider / Position)
  // 50 = Startpunkt / height-100 = max Grösse
  slider = createSlider(50, height - 100, 400);
  slider.position(width - 400, 60);
  //slider Text
  sliderText = createSlider(0, 300, 400);
  sliderText.position(width - 600, 60);

  // Button
  button = createButton('Random Background');
  button.position(positionDOM, 100);
  button.mousePressed(() => {
    bgColor = color(random(255), random(255), random(255));
  });
  bgColor = color(220);

  // ButtonText
  buttonText = createButton('Random Color Text');
  buttonText.position(positionDOM, 250);
  buttonText.mousePressed(() => {
    colorText = color(random(255), random(255), random(255));
  });
  //Start Farbe
  colorText = color(200);

  // Color Picker
  colorPicker = createColorPicker('#ff0000');
  colorPicker.position(positionDOM, 140);

  // Dropdown
  dropdown = createSelect();
  dropdown.position(positionDOM, 180);
  //Dropdown Option hinzufügen
  dropdown.option('Circle');
  dropdown.option('Square');
  dropdown.option('Circle2');
  rectMode(CENTER);

  // Input field

```

```

    input = createInput('Type text');
    input.position(positionDOM, 220);
    textAlign(CENTER, CENTER);
}

//
function draw() {
    background(bgColor);
    fill(colorPicker.value());

    if(checkbox.checked()) {
        //wenn dropdown value is circle = draw ellipse
        if(dropdown.value() === 'Circle') {
            ellipse(width / 2, height / 2, slider.value(),
slider.value());
            //wenn dropdown value is square = draw rectangle
        } else if(dropdown.value() === 'Square') {
            rect(width / 2, height / 2, slider.value(),
slider.value());
        } else if(dropdown.value() === 'Circle2') {
            ellipse(width / 2 + 100, height / 2, slider.value(),
slider.value());
            ellipse(width / 2 - 100, height / 2, slider.value(),
slider.value());
        }
    }
    //Text values von Slider und Button
    fill(colorText)
    textSize(sliderText.value());
    text(input.value(), width / 2, height / 2);
}

```


- Showed us how to create pencil tools and animations
- Experimented with snippets

```

// {"P5LIVE":{"name":"new_001","mod":1777988307010}}

let penColor = 0;

function setup() {
    createCanvas(windowWidth, windowHeight);
    background(255);
    stroke(0);
}

function draw() {

```

```

let pen1 = map(sin(frameCount * 0.005), -1, 1, 10, 50)
let differentX = map(mouseX, 0, width, 0, width / 2)

// ellipse (width/2 + cos(frameCount*0.025)*200,height/2 +
sin(frameCount*0.025)*200,100)

if(mouseIsPressed) {
  stroke(penColor);

  strokeWeight(pen1);
  line(prevX, prevY, mouseX, mouseY);
  //ellipse(mouseX,mouseY, pen1, pen1)
  //ellipse (mouseX,height/2,20)
  //ellipse (differentX,height/2-100,20)
}
prevX = mouseX;
prevY = mouseY;
//random color for pen
penColor = color((255), random(255), random(255));
}

function keyPressed() {
  if(key == 'S') {
    save('drawing.png')
  }
}

```



```

// {"P5LIVE":{"name":"colorful keypress","mod":1777990537599}}

function setup() {
  createCanvas(windowWidth, windowHeight);
  textSize(80);
  fill(255);
  background(200);
}

function draw() {
  strokeWeight(4)
  stroke(0);
  fill(random(255), random(255), random(255));
  text(key, mouseX, mouseY); // Draw at coordinate (20,75)
}

```



```
// {"P5LIVE":{"name":"random colorful circle","mod":1777993426974}}

let t = 0;

function setup() {
  createCanvas(windowWidth, windowHeight)

  setupAudio(true) // global vars
  // a5.ease = .075 // set easing
}
function draw() {
  updateAudio();

  // Noise für Bewegung über den ganzen Screen
  let x = noise(t) * width;
  let y = noise(t + 100) * height;
  t += 0.005;

  fill(random(255), random(255), random(255));
  ellipse(x, y, ampEase * 20);
}

/*
P5LIVE - Audio
If using outside P5LIVE, include p5live-audio.js
https://cdn.jsdelivr.net/gh/ffd8/P5LIVE/includes/utils/p5live-audio.js
*/
```


Day 3

- Got introduced to Hydra
- Experimented with the basics and transformations

```
osc(20, 0.1, 1.5).shift(0.4,0,0)
  .modulate(src(o0).scale(0.1), 0.15)
  .modulate(noise(3, 0.5), 0.1) // noise zu Output o0 hinzufügen
  .out(o0)
//Output 0 wird genommen und rotiert
src(o0).rotate(0.8).out(o1)
//Kaleid wird auf src01 angewendet und Scroll-Effekt wird hinzugefügt
```

```
//neuer Output heisst o2
src(o1).kaleid(10).scrollY(1, 0.2).out(o2)
;

render(o2)
```



```
osc(20, 0.1, 1.5).shift(0.4,0,0)
  .modulate(src(o0).scale(0.1), 0.15)
  .modulateRotate(noise(3, 0.5), 0.1) // noise zu Output o0 hinzufügen
  .out(o0)
//Output 0 wird genommen und rotiert
src(o0).rotate(0.8).out(o1)
//Kaleid wird auf src01 angewendet und Scroll-Effekt wird hinzugefügt
//neuer Output heisst o2 // rotate mouse.x -> kann mit maus rotiert werden
//wenn array mit eckigen kammern mehrer values hat, dann werden sie
nacheinander abgespielt
src(o1).kaleid([10,,4,50]).scrollY(1, 0.2).diff(o0).rotate( () =>
mouse.x).out(o2)
;

render(o2)
```



```
s0.initCam()
src(s0).out(o0)
src(o0).modulate(noise(7,0.5)).scale(( ) => 1 + a.fft[3]).out(o1)
render(o1)
```


- Learned how to load hydra into p5live
- Experimented with it

```
let libs = ['https://unpkg.com/hydra-synth', 'includes/libs/hydra-
synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js',
'includes/libs/hy5.js']
// sandbox - start - inside sandbox we write hydra
H.pixelDensity(2) // 2x = retina, set <= 1 if laggy
```

```

s0.initP5() // send p5 to hydra
P5.toggle(0) // optionally hide p5
//s0 = p5
src(s0)
  .mult(gradient(3))
  .modulate(noise([10,100,2]))
  //.kaleid(10,0.2)
  //.scrollY(1, 0.2)
  .brightness(0.5)
  //.scale(() => 1 + a.fft[3])
  .modulateRepeatY(noise(3))
  .pixelate(30)
  .scrollY(1,0.3)
  .out(o1)
render(o1)
// sandbox – end

function setup() {
  createCanvas(windowWidth, windowHeight)
}

function draw() {
  // clear()
  ellipse(mouseX,mouseY,200)
  noStroke()
}

```



```

/*
  _HY5_p5_hydra // cc teddavis.org 2024
  pass p5 into hydra
  docs: https://github.com/ffd8/hy5
*/

let libs = ['https://unpkg.com/hydra-synth', 'includes/libs/hydra-synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js', 'includes/libs/hy5.js']

// sandbox – start
H.pixelDensity(2) // 2x = retina, set <= 1 if laggy

s0.initP5() // send p5 to hydra
P5.toggle(0) // optionally hide p5
//Hintergrund erstellen
osc(107, 0, 0.7)

```

```

    .color(0.4, 1, 1)
    .rotate(0, -0.08)
    .modulateRotate(o1, 0.4)
    .out(o0)
//src o0 nehmen und verändern und dann Maske mit Form hinzufügen
src(o0)
    .add(osc(33))
    .modulateRepeat(noise(3))
    .diff(o0)
//Zwei Dreiecke ergeben Stern
    .mask(shape(3).add(shape(3).rotate(1.3)).rotate(0, 0.08))
    .out(o1)
src(o1).osc(20,10,4).out(o2)

// sandbox – end

function setup() {
    createCanvas(windowWidth, windowHeight)
}

```



```

// {"P5LIVE":{"name":"pixel text audio","mod":1778154280316}}

/*
    _HY5_p5_hydra // cc teddavis.org 2024
    pass p5 into hydra
    docs: https://github.com/ffd8/hy5
*/

let libs = ['https://unpkg.com/hydra-synth', 'includes/libs/hydra-synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js', 'includes/libs/hy5.js']

// sandbox – start
H.pixelDensity(2) // 2x = retina, set <= 1 if laggy

s0.initP5() // send p5 to hydra
P5.toggle(0) // optionally hide p5

src(s0)

```

```

    .modulate(noize(2,1))
    .scrollY(1, 1.5)
    .pixelate(150)
    // .scale(() => 0.01 + a.fft[0])
    .kaleid([1,50,5])

    .out()
// sandbox - end

function setup() {
  createCanvas(windowWidth, windowHeight)
}
function draw() {
  background(20,30,255)
  textAlign(CENTER,CENTER)
  fill(255,0,100)
  textSize(400)
  text('erdbeeri',width/2,height/2)
  fill(0,255,0)
  text('mango',width/2-200,height/2+200)
  fill(255,255,0)
  text('kiwi',width/2+200,height/2-200)
}

```


Day 4

- Jasmin showed us some of her work
- Learned how we can edit and use Text in p5live
- Experimented with the different Text options

```

/*
  _HY5_p5_hydra // cc teddavis.org 2024
  pass p5 into hydra
  docs: https://github.com/ffd8/hy5
*/

let libs = ['https://unpkg.com/hydra-synth', 'includes/libs/hydra-synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js', 'includes/libs/hy5.js']

// sandbox - start
H.pixelDensity(2) // 2x = retina, set <= 1 if laggy

```

```

s0.initP5() // send p5 to hydra
P5.toggle(0) // optionally hide p5

src(s0)
  .modulate(noize(2,1))
  .scrollY(1, 1.5)
  .pixelate(150)
  //.scale(() => 0.01 + a.fft[0])
  .kaleid([1,50,5])

  .out()
// sandbox – end

function setup() {
  createCanvas(windowWidth, windowHeight)
}

function draw() {
  background(20,30,255)
  textAlign(CENTER,CENTER)
  fill(255,0,100)
  textSize(400)
  text('erdbeeri',width/2,height/2)
  fill(0,255,0)
  text('mango',width/2-200,height/2+200)
  fill(255,255,0)
  text('kiwi',width/2+200,height/2-200)
}

```



```

function setup() {
  createCanvas(windowWidth, windowHeight)
}

function draw() {
  let live = frameCount%10
  //Damit Framecount weniger schnell ist
  frameRate(7)
  background(0,0,255)
  fill(255)
  textSize(100*(live/2))
  //Zeilenumbruch bei Wort oder Character
  textWrap(CHAR)
  textFont('monospace')
}

```

```

    textAlign(LEFT)
    textStyle(random([NORMAL,ITALIC]))
    //Zeilenabstand = textLeading, animiert mit Framecount
    textLeading(50*(live/2))
    //.repeat um Text zu wiederholen
    text("zürich zürich ".repeat(100), 100,100,
    windowWidth,windowHeight)
}

```



```

function setup() {
  createCanvas(windowWidth, windowHeight)
}

function draw() {
  let live = frameCount%10
  let words = ["-- --", "ooo oo", ",,,, ", "!!! !!", "???? ?", "<< <<<", "...", "||| |||||", "xx x", ">> >>>"];
  let rand = random(words);
  let sine = floor(5*sin(frameCount/10)+5)
  //Damit Framecount weniger schnell ist
  frameRate(10)
  background(0,0,255)
  fill(255)
  textSize(200)
  //Zeilenumbruch bei Wort oder Character
  textWrap(CHAR)
  textFont('monospace')
  textAlign(LEFT)
  textStyle(NORMAL)
  //Zeilenabstand = textLeading, animiert mit Framecount
  textLeading(30*(live/2))
  //.repeat um Text zu wiederholen
  text(words[sine].repeat(1000), 100,100,
  windowWidth-100,windowHeight-100)
}

```



```

function setup() {
  createCanvas(windowWidth, windowHeight)
}

```

```

function draw() {
  let live = frameCount%10
  let words = ["-- --", "ooo oo", ",,,, ", "!!! !!", "???? ?", "<< <<<", "...", "||| |||||", "xx x", ">> >>>"];
  let rand = random(words);
  let sine = floor(5*sin(frameCount/10)+5)
  //Damit Framecount weniger schnell ist
  frameRate(10)
  background(0,0,255)
  fill(255)
  textSize(200)
  //Zeilenumbruch bei Wort oder Character
  textWrap(CHAR)
  textFont('monospace')
  textAlign(LEFT)
  textStyle(NORMAL)
  //Zeilenabstand = textLeading, animiert mit Framecount
  textLeading(30*(live/2))
  //.repeat um Text zu wiederholen
  text(words[sine].repeat(1000), 100,100,
  windowWidth-100,windowHeight-100)
}

```



```

let libs = ['https://unpkg.com/hydra-synth', 'includes/libs/hydra-synth.js', 'https://cdn.jsdelivr.net/gh/ffd8/hy5@main/hy5.js', 'includes/libs/hy5.js']

// sandbox - start
H.pixelDensity(2) // 2x = retina, set <= 1 if laggy

s0.initP5() // send p5 to hydra
P5.toggle(0) // optionally hide p5

src(s0)
  .rotate(10)
  .out()
// sandbox - end
function setup() {
  createCanvas(windowWidth, windowHeight)

  // setupAudio(true) // global vars
  // a5.ease = .075 // set easing
}

function draw() {

```

```

/* audio vars: amp, ampEase, fft, fftEase, waveform, waveformEase */
//updateAudio()
    let live = frameCount%10
    let words = ["-- --","jjj jj","hässig ","222 22","o00o 00","dunnstig
",".. ...","namittag ","xx x","%% %%"];
    let rand = random(words);
    let sine = floor(5*sin(frameCount/10)+5)
    //Damit Framecount weniger schnell ist
    frameRate(5)
    background(127,255,0)
    fill(0)
    textSize(100)
    //Zeilenumbruch bei Wort oder Character
    textWrap(CHAR)
    textFont('monospace')
    textAlign(LEFT)
    textStyle(random( [NORMAL,ITALIC]))
    //Zeilenabstand = textLeading, animiert mit Framecount
    textLeading(100)
    //.repeat um Text zu wiederholen
    text(words[sine].replace(/ /g,"!!!").repeat(1000), 100,100,
    windowWidth-200,windowHeight-200)

}

```



```

function setup() {
    createCanvas(windowWidth, windowHeight)
}

function draw() {
    let live = frameCount%10
    let words = "hässig ";
    let rand = random(words);
    let sine = floor(5*sin(frameCount/10)+5)
    //Damit Framecount weniger schnell ist
    frameRate(10)
    background(0,0,255)
    fill(255)
    textSize(100)
    //Zeilenumbruch bei Wort oder Character
    textWrap(CHAR)
    textFont('monospace')
    textAlign(LEFT)

```

```

textStyle(ITALIC)

//Zeilenabstand = textLeading, animiert mit Framecount
textLeading(30*(live/2))
//.replace(/x/g,"a") = alle x werden mit a ersetzt
text(words.replace(/i/g,"!").repeat(1000), 500,100,
windowWidth>windowHeight)
}

```



```

function setup() {
  createCanvas(windowWidth, windowHeight)
  frameRate(10)
  noSmooth()
}

function draw() {
  background(30)
  let comma = ","
  let space = ["-"]
  let commaline;
  fill(255)
  textSize(windowWidth/5)
  textWrap(CHAR)
  textSize(windowWidth/50)
  textFont("monospace")
  //loop wiederholen bis i nicht mehr kleiner als 11 ist
  for (let i = 0; i < 11; i++){
    space.push("----")
    // space[i] wird bei jeder runde an comma angehängt = wird immer
    länger
    comma = comma + space[i]
    //wiederholt es 33 mal
    commaline = comma.repeat(33)
    // 100 = Abstand vom links
    //commaline.length*i/1.1 = macht dass line immer weiter unten ist,
    desto länger sie wird
    text(commaline,100,commaline.length*i/1.1,
windowWidth/2.5>windowHeight);
  }
}

```

